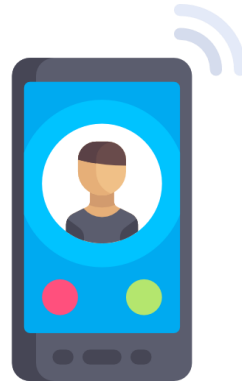


1. Introduction to Mobile Application Development



Objectives

Know current state of mobile application development

Explain the Android Application Fundamentals

Explain components of an Android app

Explain the following terms :

Activities

Broadcast Receivers

Services

Intent

Content Providers

Manifest File

The current state of mobile application development

The current state

Types of mobile apps

Programming languages and Tools

Mobile operating systems

Key challenges

Beyond mobile app

Types of Mobile Apps

Native



Hybrid

Web
+
Native

Web



Comparison of Apps

	Native	Hybrid	Mobile Web
Cost	High	Low	Low
Performance	Fast	Depending on speed of network	Depending on speed of network
Distribution	App stores	App stores	None
Device features	Wide	Limited	Very limited
Code maintenance	Multiple codebase	Single codebase	Single codebase

Question?

According to Medium (2019), the four market moving mobile app trends in 2019 are:

1. App store consumer spending will surpass USD 120 billion globally
2. Mobile gaming market share will increase to 60%
3. Users will spend 10 minutes of every hour consuming streaming video on a mobile device
4. Publishers will produce 60% more apps with in-app ads, making up 62% of the USD 250 billion global digital ad spend

In your opinion, which mobile app development method (native, mobile web, and hybrid) is suitable for each the following entities?

- a. Bernama, the Malaysian national news-agency
- b. Super Mario Bros, a game developed by Sega
- c. Setapak Central, a shopping mall

Programming Languages and Tools

Programming Languages

Native



Hybrid

Web
+
Native

Web



Development Tools - Native



Android Studio



XCode



Tizen Studio

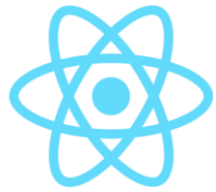


Visual Studio



Huawei Quick App

Development Tools – Cross Platform



React Native



Flutter



Xamarin



PhoneGap



Apache Weex



Native Script

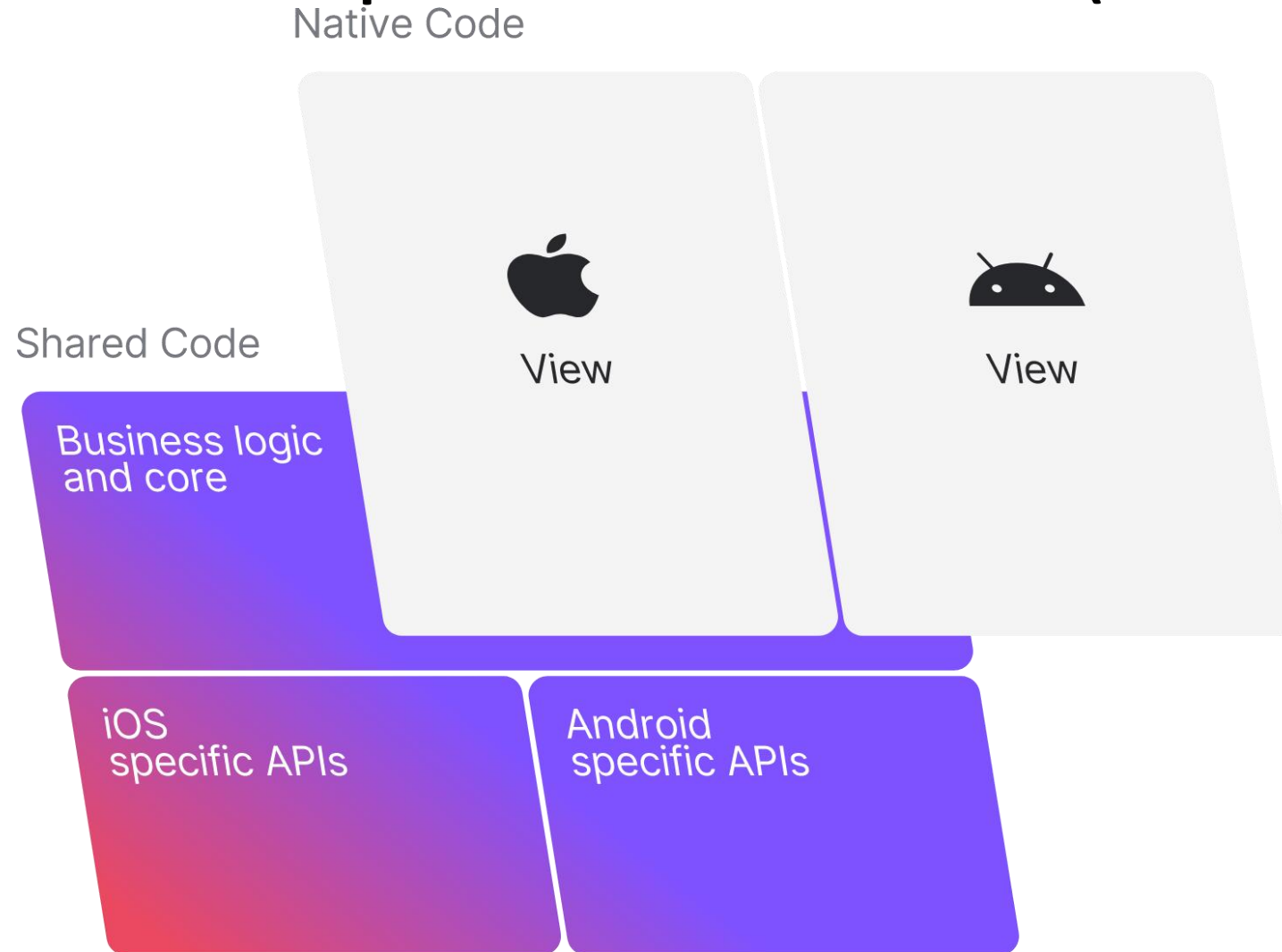


Jasonette



Ionic

Kotlin Multiplatform Mobile (KMM)

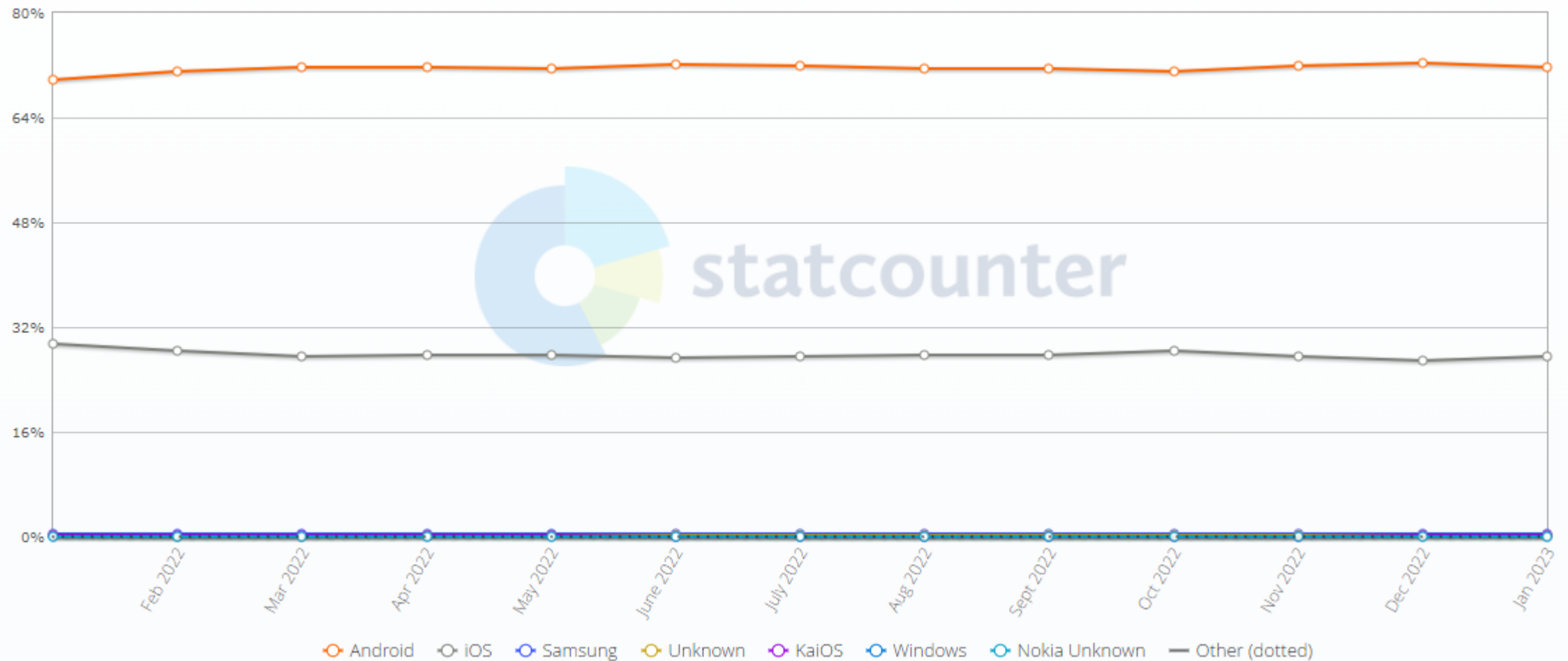


Mobile Operating Systems

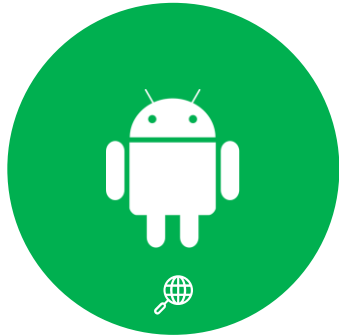
Mobile OS

Mobile Operating System Market Share Worldwide

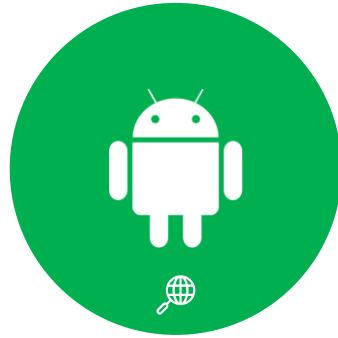
Jan 2022 - Jan 2023

[Edit Chart Data](#)

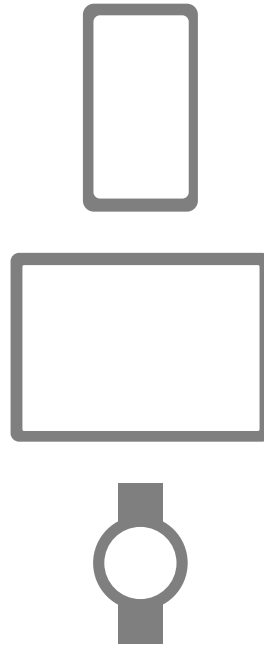
Mobile OS



Mobile OS - Android



Linux kernel
Open Source

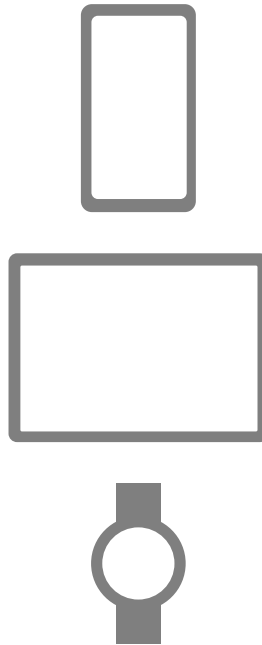


Chrome OS
Linux-based
Open Source

Mobile OS - iOS



Unix kernel



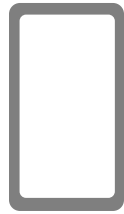
macOS

Mobile OS - Tizen



Linux kernel

Open Source



Mobile OS - KaiOS



Linux kernel

Open Source

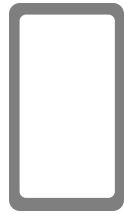


Mobile OS – Harmony OS



Microkernel

Open Source



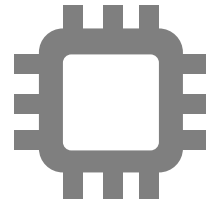
Mobile OS - Fuchsia



Microkernel

Open Source

Module Architecture



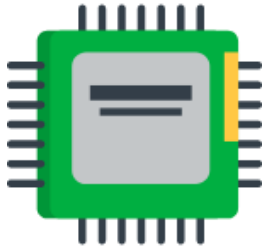
ARM-based



Nest Hub

Key Mobile Challenges

Key Mobile Challenges



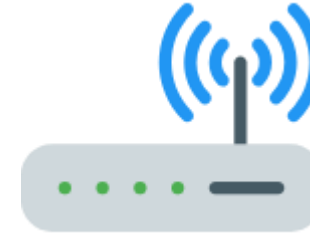
Processing Power



Memory and
storage



Battery



Network

Key Mobile Challenges



Flagship



Mid Range



Entry



Basic



Octa-core (SD 888)
Hexa-core (A12)



Quad-core



12 GB RAM / 256 GB ROM



2 GB RAM / 8 GB ROM



6000 mAh



2150 mAh



4G/5G Network



3G/4G Network

Question?

India, a country with a large population base, low smartphone adoption and fast growth rate. Mobile phone shipments in the Indian market already exceed those in the US, making it the second largest phone market. The current Indian mobile phone market is still in the first batch of smartphone replacement. The problem for many people there is to have one (Sina Tech, 2018).

Considering the four challenges: processing power, memory and storage, battery, and network, how would you build apps to cater to the needs of users in emerging markets?

[Sina Tech \(2018\), available online:https://pandaily.com/oppo-vivo-cut-retail-margins-40-lose-10000-outlets/](https://pandaily.com/oppo-vivo-cut-retail-margins-40-lose-10000-outlets/)

Beyond Mobile App

Beyond Mobile App



Wearable



Smart Home Appliance



Smart Car Dashboard



Internet of Things (IoT)

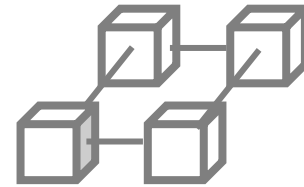
Beyond Mobile App



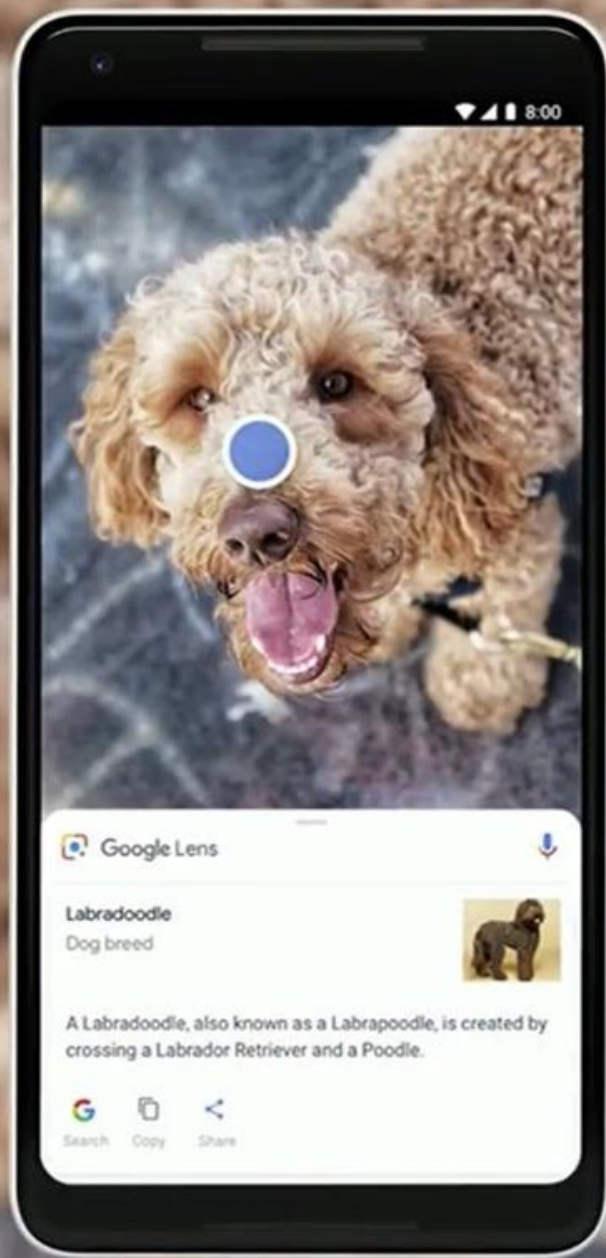
Augmented Reality (AR)



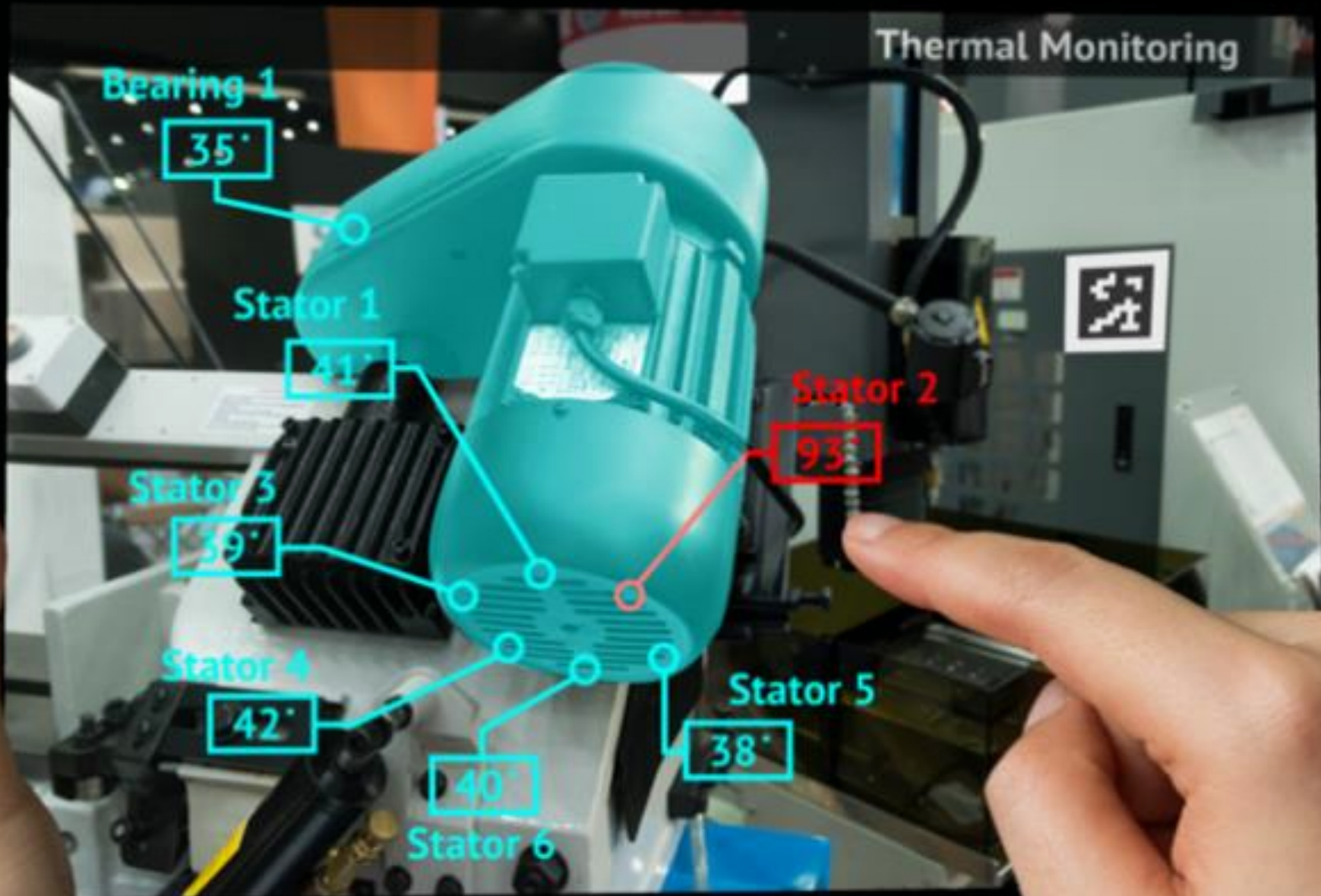
Artificial Intelligent (AI)

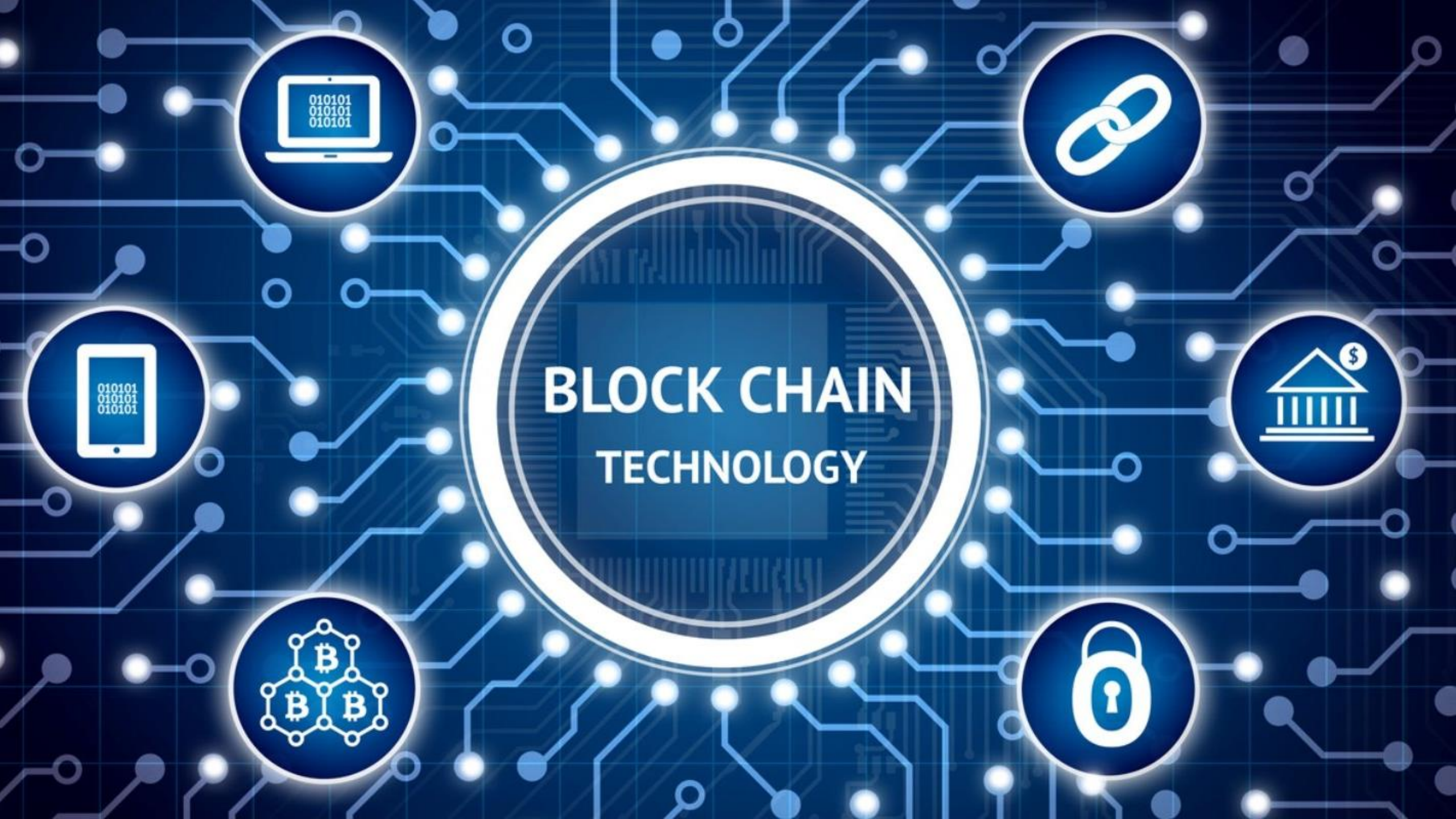


Blockchain



Thermal Monitoring





A conceptual graphic for blockchain technology. The background is dark blue with a network of glowing white lines and dots. A large central circle contains the text "BLOCK CHAIN TECHNOLOGY". Surrounding this central circle are six smaller circles, each containing a different icon: a laptop with binary code, a chain link, a bank building with a dollar sign, a padlock, a Bitcoin network diagram, and a tablet with binary code.

BLOCK CHAIN TECHNOLOGY





Android Application Fundamentals

App Components

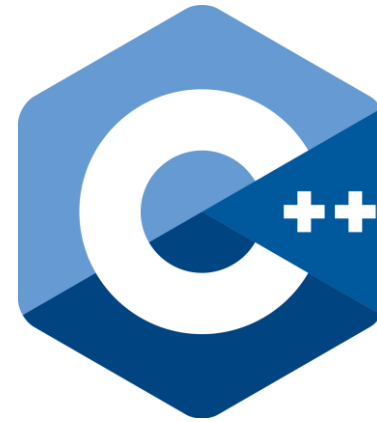
- Android apps are written in:



Java

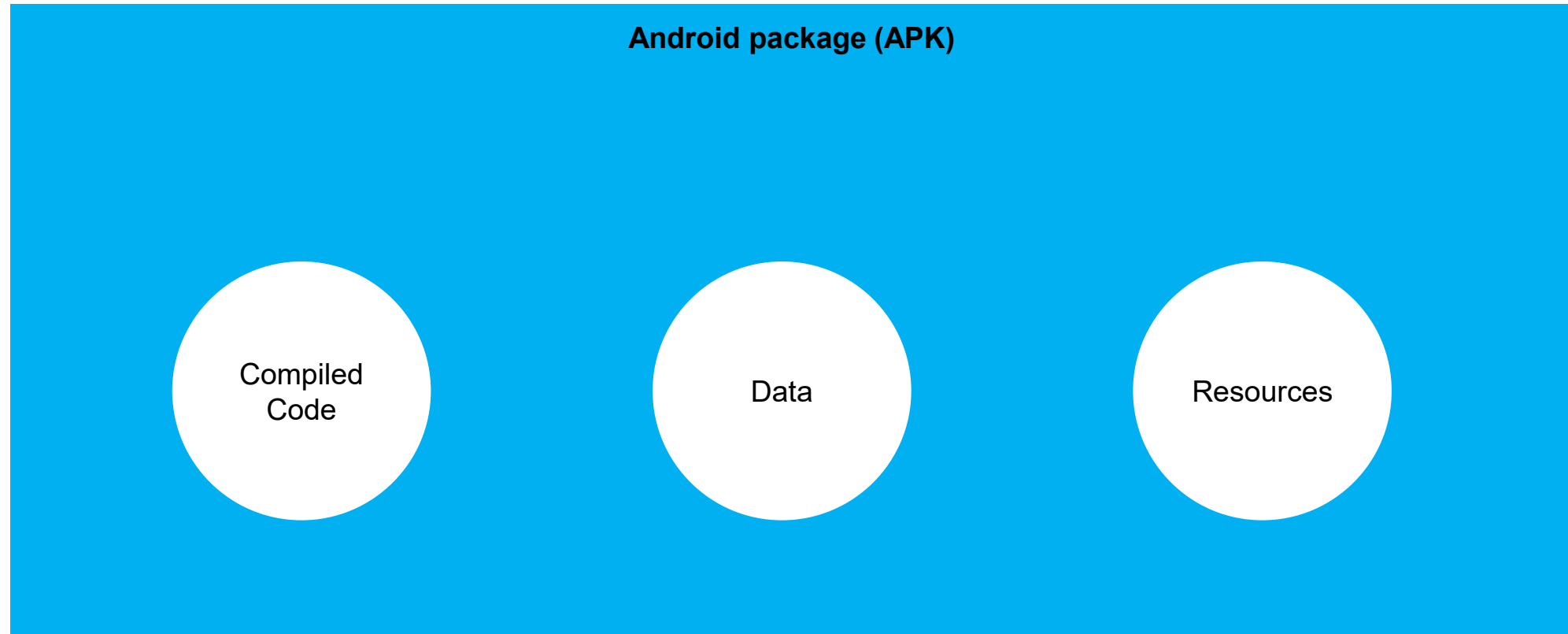


Kotlin



C++

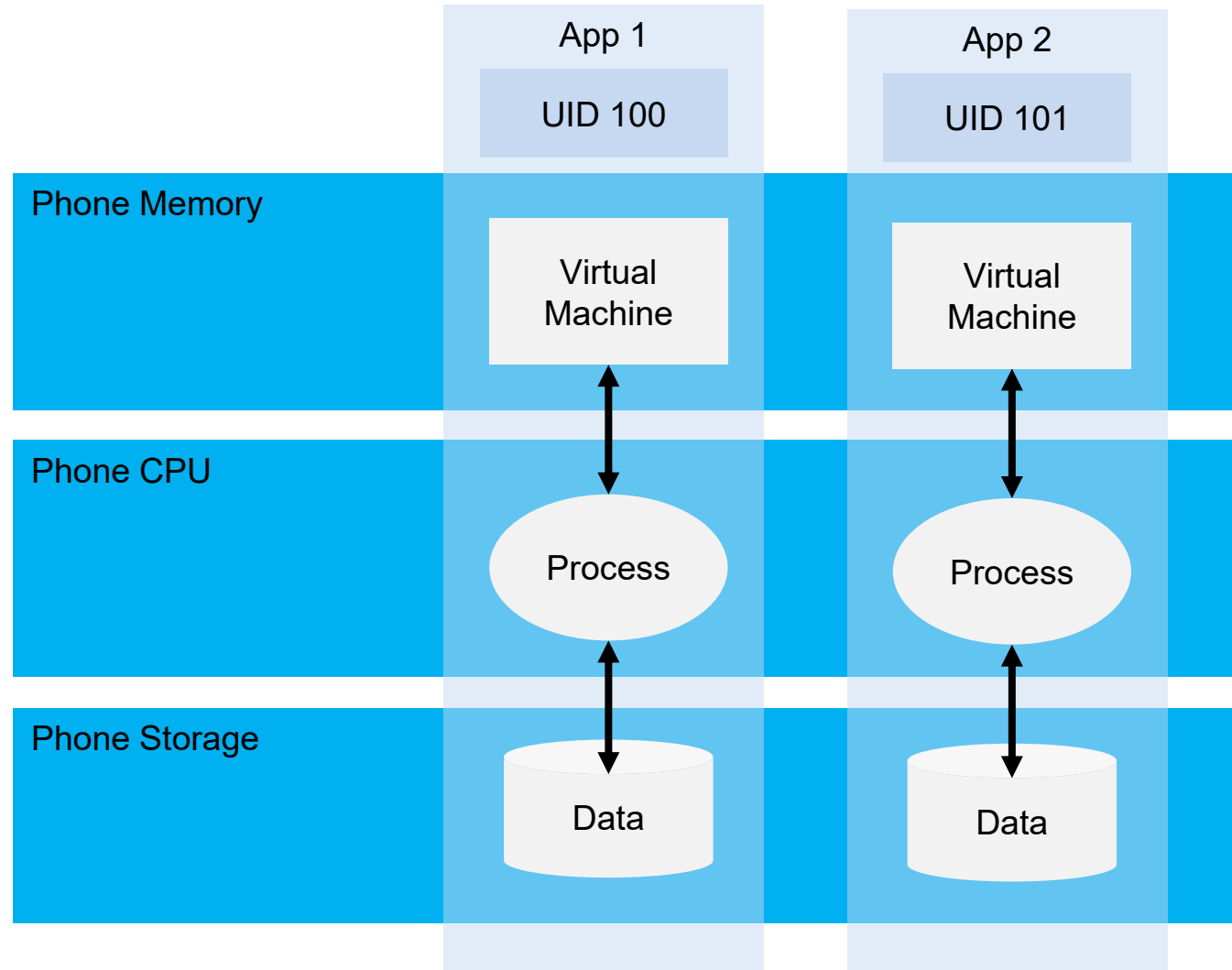
App Components



Android Application Fundamentals

- Each Android app lives in its own security sandbox
 - Each app is a different user to the OS
 - The system assigns each app a unique Linux ID
 - The system sets permission for all the files in an app
 - Each app runs in its own Linux process
 - Each process has its own virtual machine (VM)

Android Security



Android Application Fundamentals

The Principle of Least Privilege

Each app, by default, has access only to the components that it requires to do its work and no more

An app cannot access parts of the system for which it is not given permission

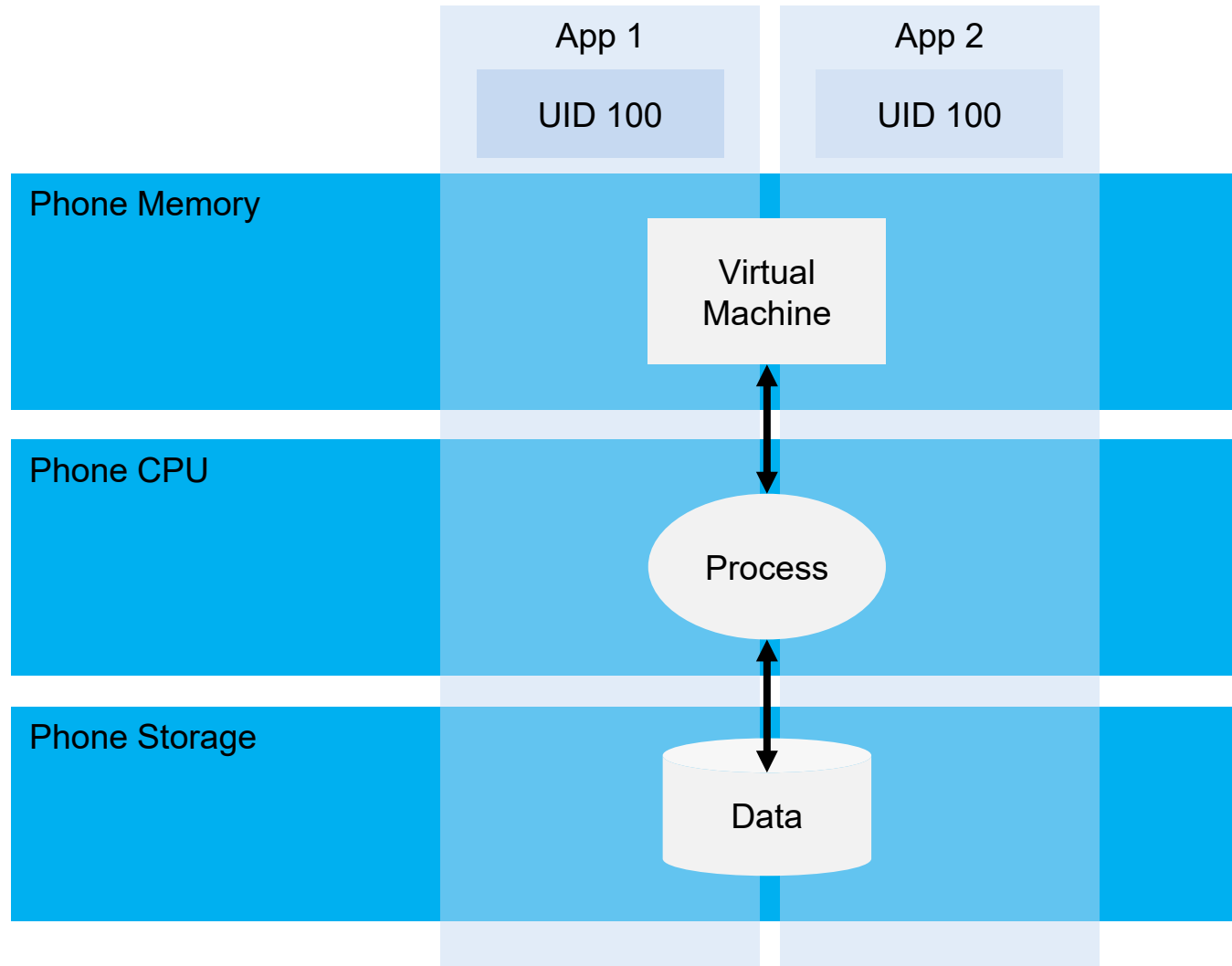
Android Application Fundamentals

To share data

- Two apps could share the same Linux user ID to allow access each other's files
- The two apps must be signed with the same certificate

Reason: to conserve system resources; apps run in the same Linux process and share VM

Two apps with the same ID



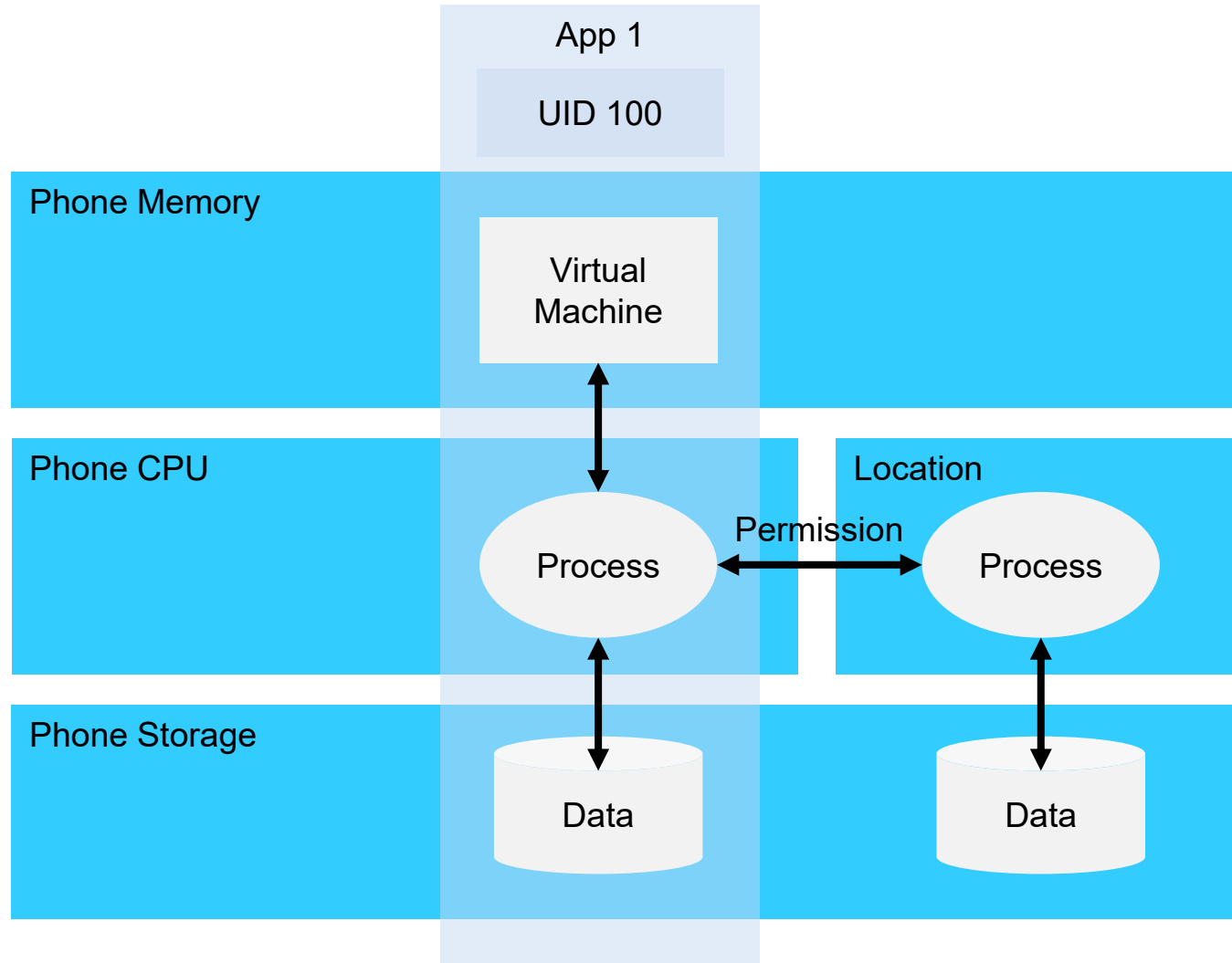
Android Application Fundamentals

To access system services

An app can request permission from a user to access device data. E.g. contact, SMS, storage, camera, Bluetooth, and etc

All app permissions must be granted by the user at install time (version 5 and below) or during runtime (version 6 and above)

App to access system service

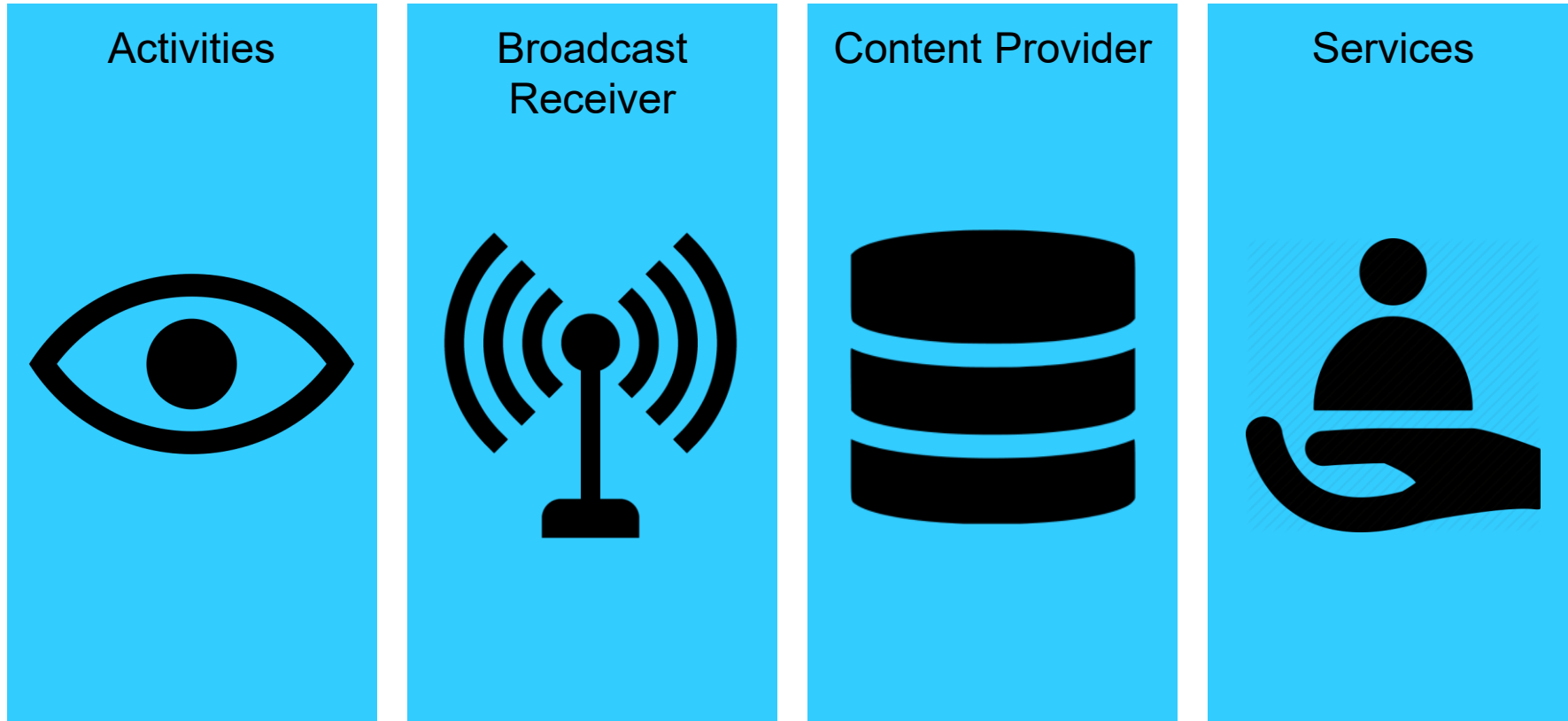


Question?

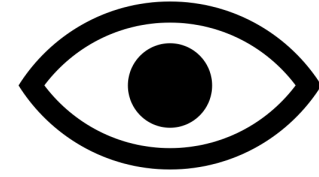
1. Why Android's application package (APK) file contains three different components?
2. Explain two ways the Android applied to protect app data and user's security

App Components

Four types of app components: ABCS



Activities



An activity = a screen

A window in which to draw its user interface (UI)

An app may consist of multiple activities

One of the activity is the “main” activity that launching an app

Broadcast Receiver



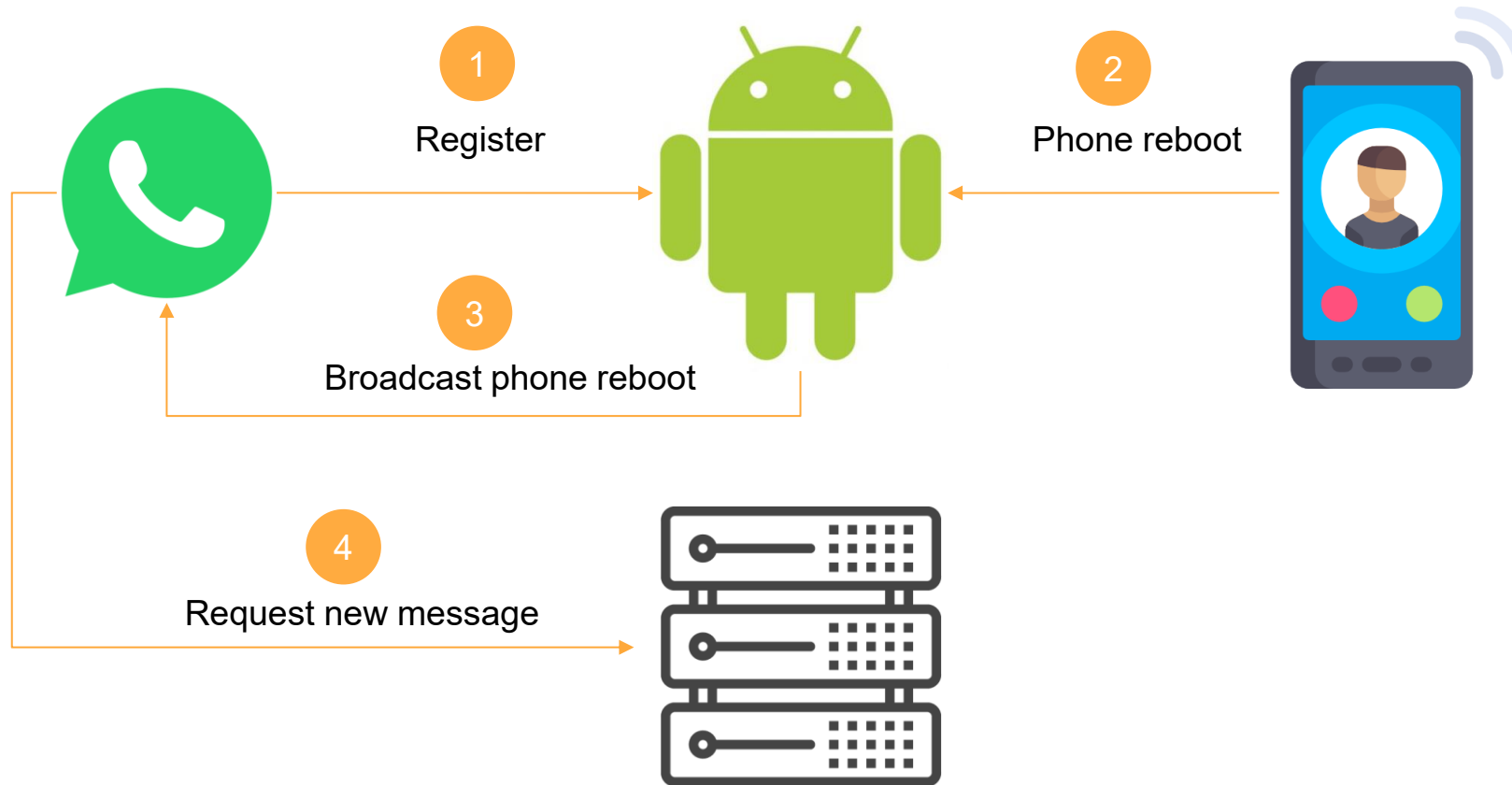
A messaging system used by the system and apps

No UI, but could create status bar notification

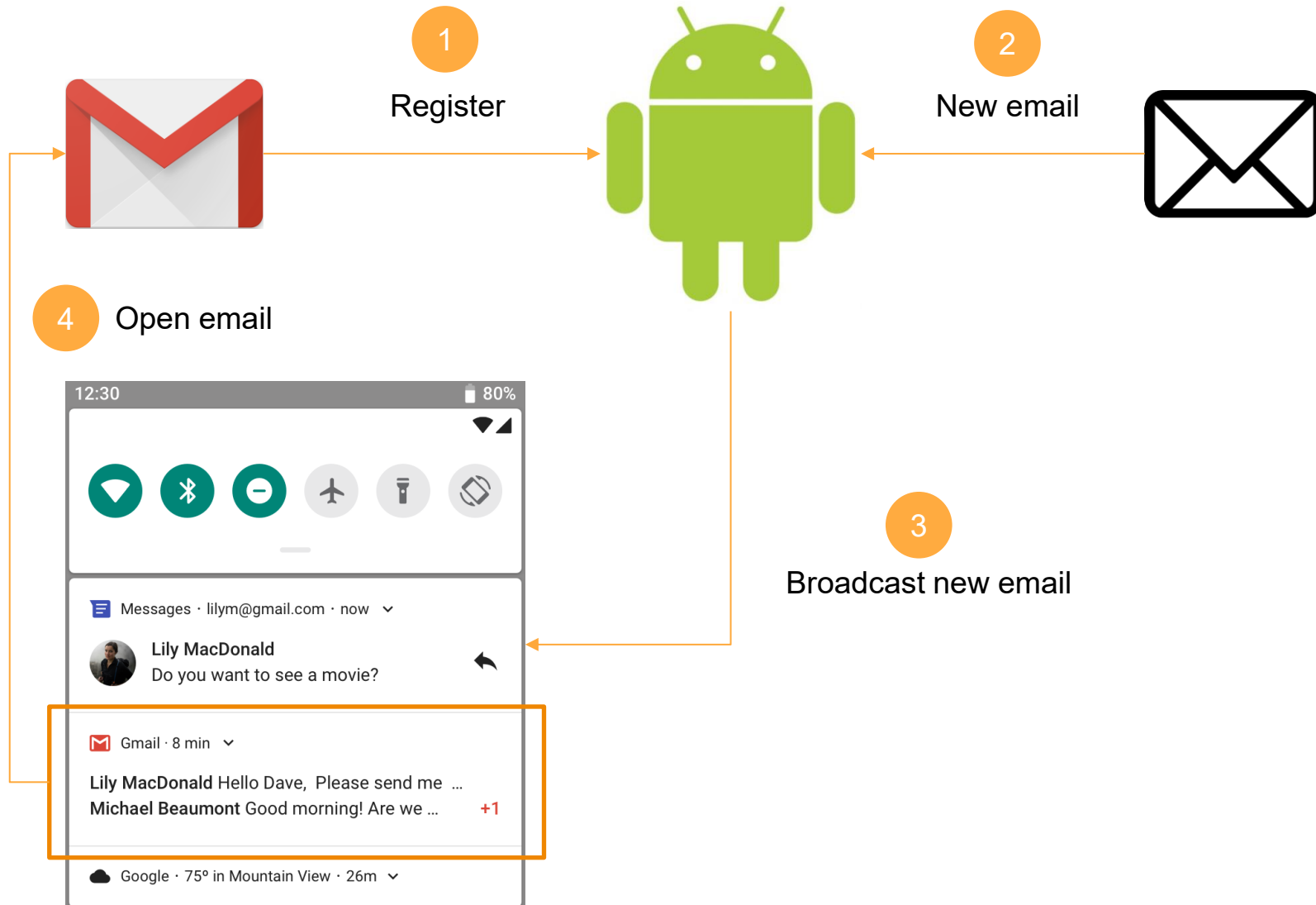
An app must register to receive an event

A system broadcast is sent to all apps that subscribed to receive the event. E.g. battery low, system reboot

Broadcast Receiver



Broadcast Receiver



Content Providers



Manages access to a central repository of data

Data formats:

- File system

- SQLite database

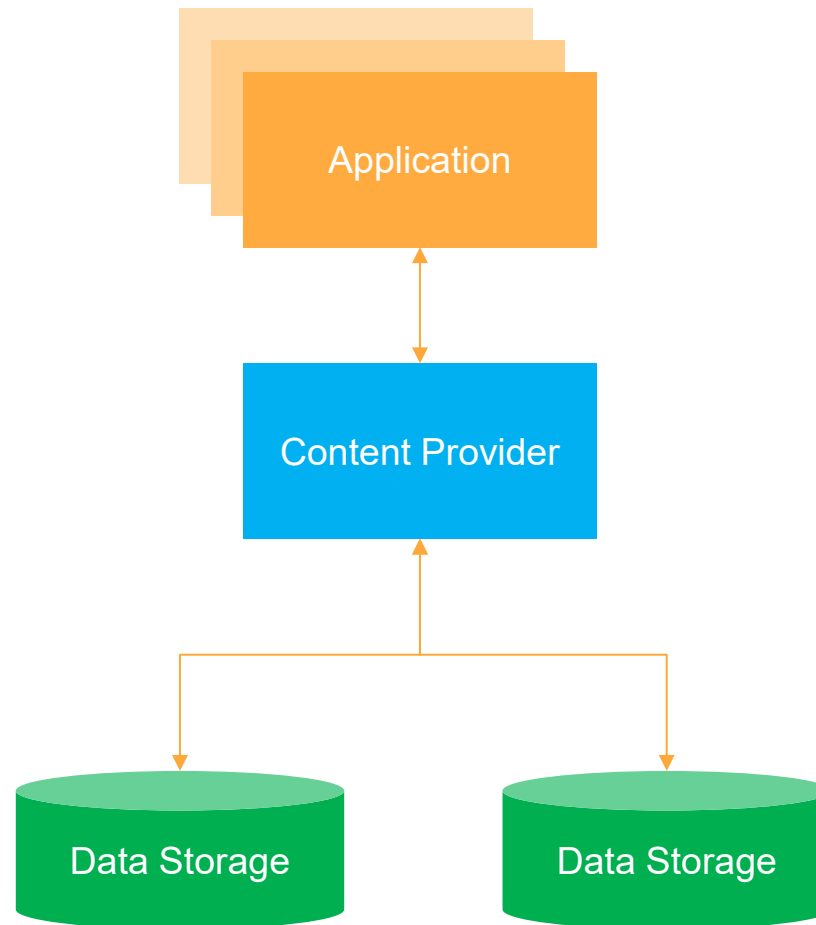
- On the web/ cloud

- Persistent storage location

Other apps can query or even modify the data (with permission)

- E.g. User dictionary

Contents Providers



Services



No UI. Runs in the background

Performs long-running operations or works for remote processes.

E.g. play music in the background or fetch data over network

Another component can start the service

Question?

Identify a type of app component suitable for each of the following:

- a. Login screen
- b. Sync data from an app to a server
- c. Manage a shared set of app data on a local storage
- d. Schedule an alarm to post a notification

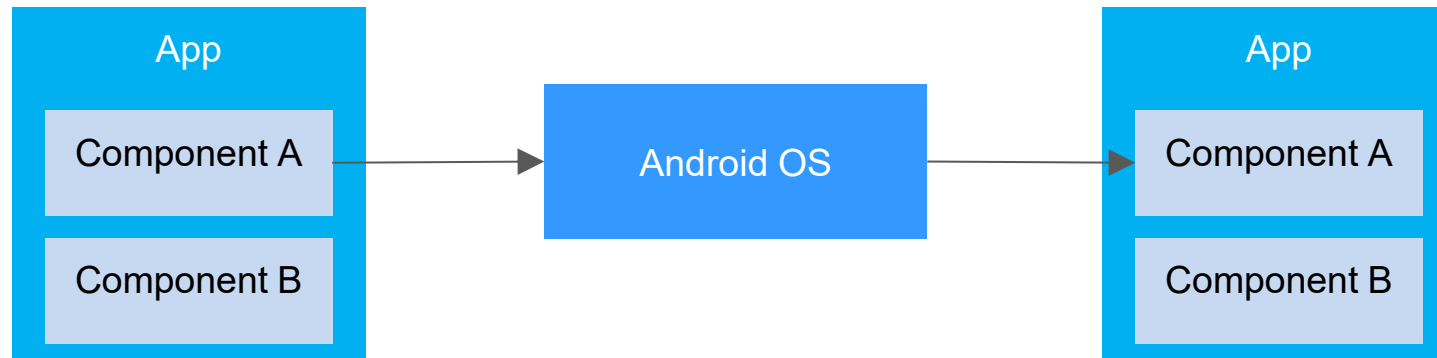
App Components

Android system runs each app in a separate process with file permissions that restrict access to other apps, how can an app activate a component from another app?

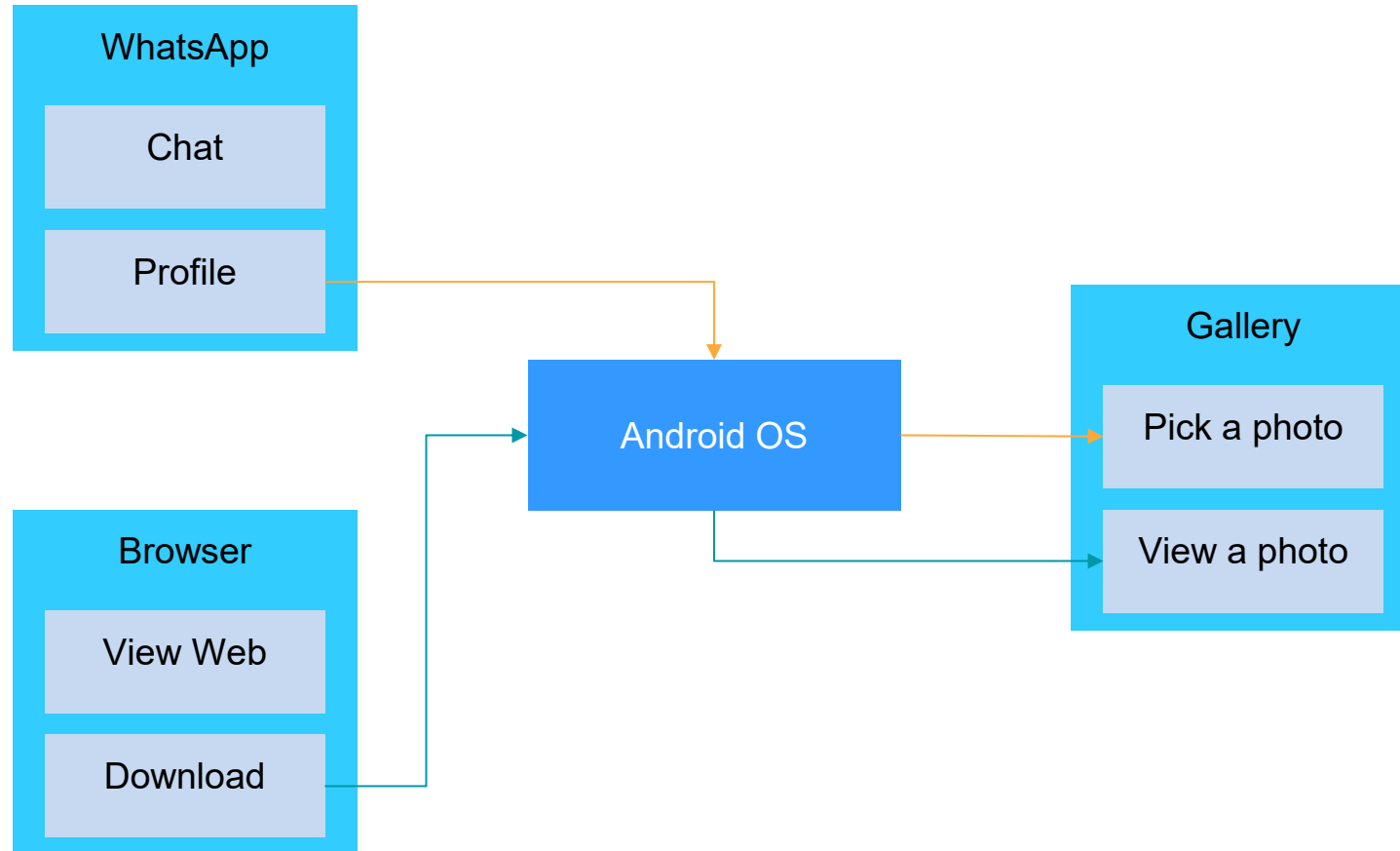
App Components

An app can start another app's component

An app does not have a single entry point; no `main()` function



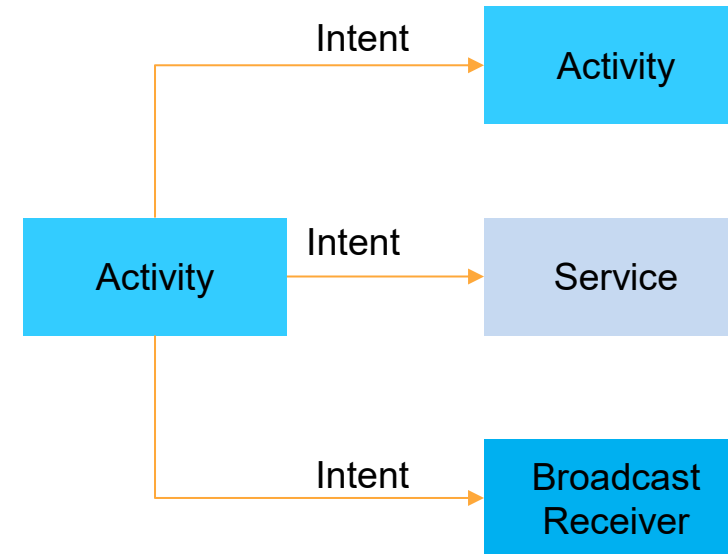
App Components



Intent

Use Intent to activate Activities, Services, and Broadcast Receivers

Intent defines a message to activate a specific component or type of component (will be discussed in Chapter 3)



The Manifest File

The first file read by the Android before it starts an app component

To know the components exists of an app

Components NOT declared in the manifest file are NOT visible to Android; can NEVER run

The Manifest File

Contains:

- Permission
- Minimum App Programming Interface (API) level
- Hardware and software features used
- App components
- Linked libraries
- etc

The Manifest File

Permission :

- A restriction limiting access to:
 - code
 - data on the device
- Protects critical data and code from misuse or damage

App Resources

Contents:

- Animation/Image
- Menu
- Style
- Colour
- String
- Layout (UI)
- etc

Question?

1. What is the purpose of the Intent class?
2. Describe three common use cases for using an Intent.
3. Why is the manifest file important to every Android app?

Conclusion

Know current state of mobile application development

Explain the Android Application Fundamentals

Explain components of an Android app

Explain the following terms :

Activities, Services, Content Providers, Broadcast Receivers, Intent, and Manifest File